

Tahlia Yue Wu

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EDUCATION

University of Pennsylvania, Philadelphia, PA - GPA: 4.0/4.0 (BSE), 4.0/4.0 (MSE)

May 2028

Candidate for BSE in Mechanical Engineering, MSE in Robotics

Relevant Courses: Machine Design and Manufacturing, Design of Mechatronic Systems, Thermodynamics

SKILLS

Software: Python, C, C++, Java, R, HTML/CSS, LabVIEW, Git, Github

Robotics & Controls: PID Control, Closed-Loop Feedback, Kinematics, OptiTrack Motion Capture, Raspberry Pi, Arduino

CAD & Manufacturing: SolidWorks (CSWA Certified), Mastercam, CNC Machining, Manual Mills & Lathes

Data & Modeling: PyTorch, Computer Vision (3D Landmark Identification), EPA MOVES, RLINE

RESEARCH EXPERIENCE

Research Assistant

Jan 2025-Present

Aware Lab, University of Pennsylvania, Philadelphia, PA

- End-to-end engineered the FlowBot, a 3D robotic traverse powered by a Raspberry Pi (Python software) and a mecanum chassis driven by DC motors, automating 30+ hours of logging anemometry data
- Achieved ~6mm spatial positioning accuracy on the FlowBot by implementing PID closed-loop feedback control integrated with an OptiTrack infrared motion capture system
- Designed and built a custom 4-foot vertical linear actuator from scratch on a <\$200 budget
- Architected wireless LabVIEW communication to integrate the FlowBot into a larger wind turbine control software program

Research Intern - NSF REU Program

May 2025-Jul 2025

Department of Civil, Environmental and Geodetic Engineering, Ohio State University, Columbus, OH

- Engineered a computational framework linking Streetlight data with EPA MOVES and RLINE to accelerate air pollutant simulations
- Computed near-road pollution distributions for 12 pollutants across a 36-road network using R
- Presented air quality research findings at the Ohio State University Summer Symposium

AI Research Intern

Jun 2023-Oct 2023

Davis Lab of Computer Vision, UC Santa Cruz, Santa Cruz, CA

- Engineered a C & Python algorithm to automate labelling of 468 facial landmarks in 3D facial point maps to boost computer vision AI model accuracy
- Software tested a PyTorch machine learning model to predict 3D facial geometry from 2D images
- Led a team of 3 lab interns, streamlining workflows and accelerating research collaboration

Data Analyst Intern

Jun 2022-Jul 2024

James Lab, Stanford University, CA

- Programmed Python-based software that automates biodistribution PET data analysis, reducing processing time from 1.5 hours to 20 minutes

WORK EXPERIENCE

Calculus Course Grader

Aug 2026-Present

University of Pennsylvania, PA

- Evaluated and graded 20+ weekly problem sets on analytical ODE solution methods, Fourier series, and boundary-value PDEs

Residential Advisor

Aug 2026-Present

Lauder College House, University of Pennsylvania, PA

- Organized 5-8 monthly community-building, cultural, and wellness events for 350 residential students.
- Partnered with college house senior staff and student peers to design and execute inclusive community programming

PUBLICATIONS & PRESENTATIONS

Speaker "Robotic Flow Measurements of Unsteady Wind-Turbine Wakes" (APS Division of Fluid Dynamics, 2026)

2nd-Author "A scalable framework for spatiotemporal modeling of traffic-related emissions" (Submitted to ACS ES&T Air, 2026)

Co-Author "Noninvasive Imaging of Myeloid Cell Dynamics: GPR84 PET Detects EAE Pathology and Tracks Therapy Response" (Published in Journal of Neuroinflammation, 2026)

AWARDS

(2025) National Science Foundation Research Experience for Undergraduates Recipient

(2026) UPenn-SEAS Advancing Women in Engineering Grant Recipient